EXPERIMENT : 11

Performance Evaluation of OFDM and FBMC Wireless Waveforms

OBJECTIVES :

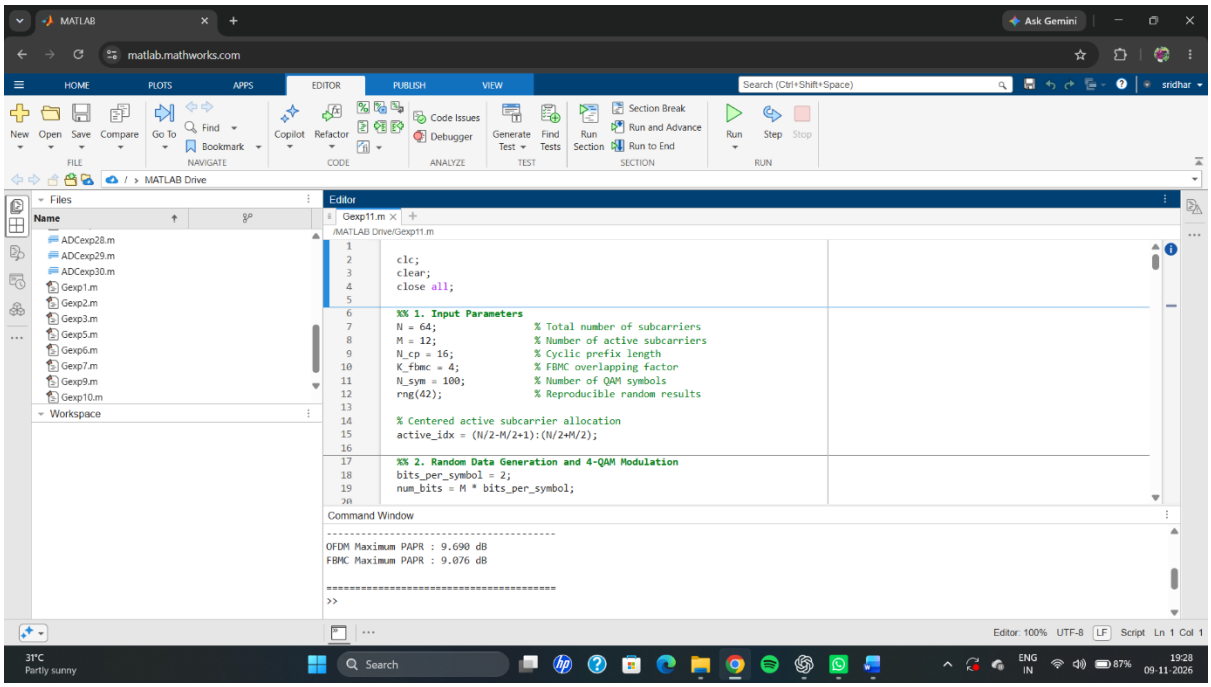
1. To generate a basic Orthogonal Frequency Division Multiplexing (OFDM) signal and a Filter Bank Multicarrier (FBMC) signal in MATLAB.

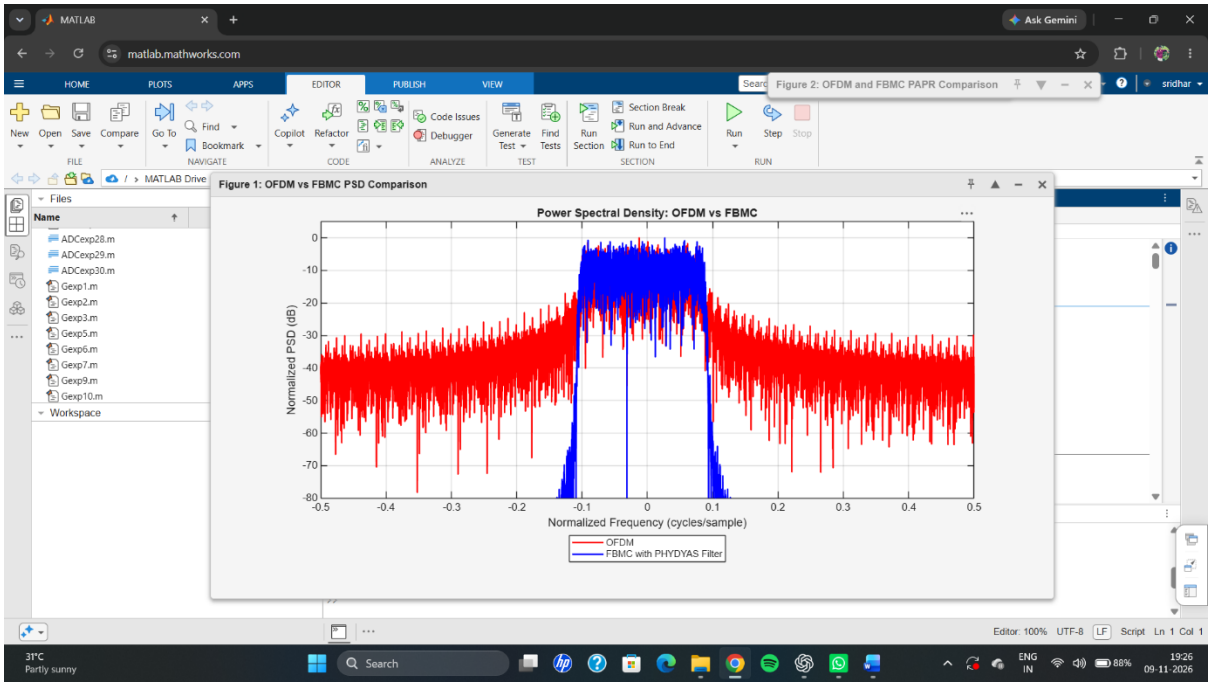
2. To compare their spectral characteristics by plotting and analyzing their power spectral density (PSD).

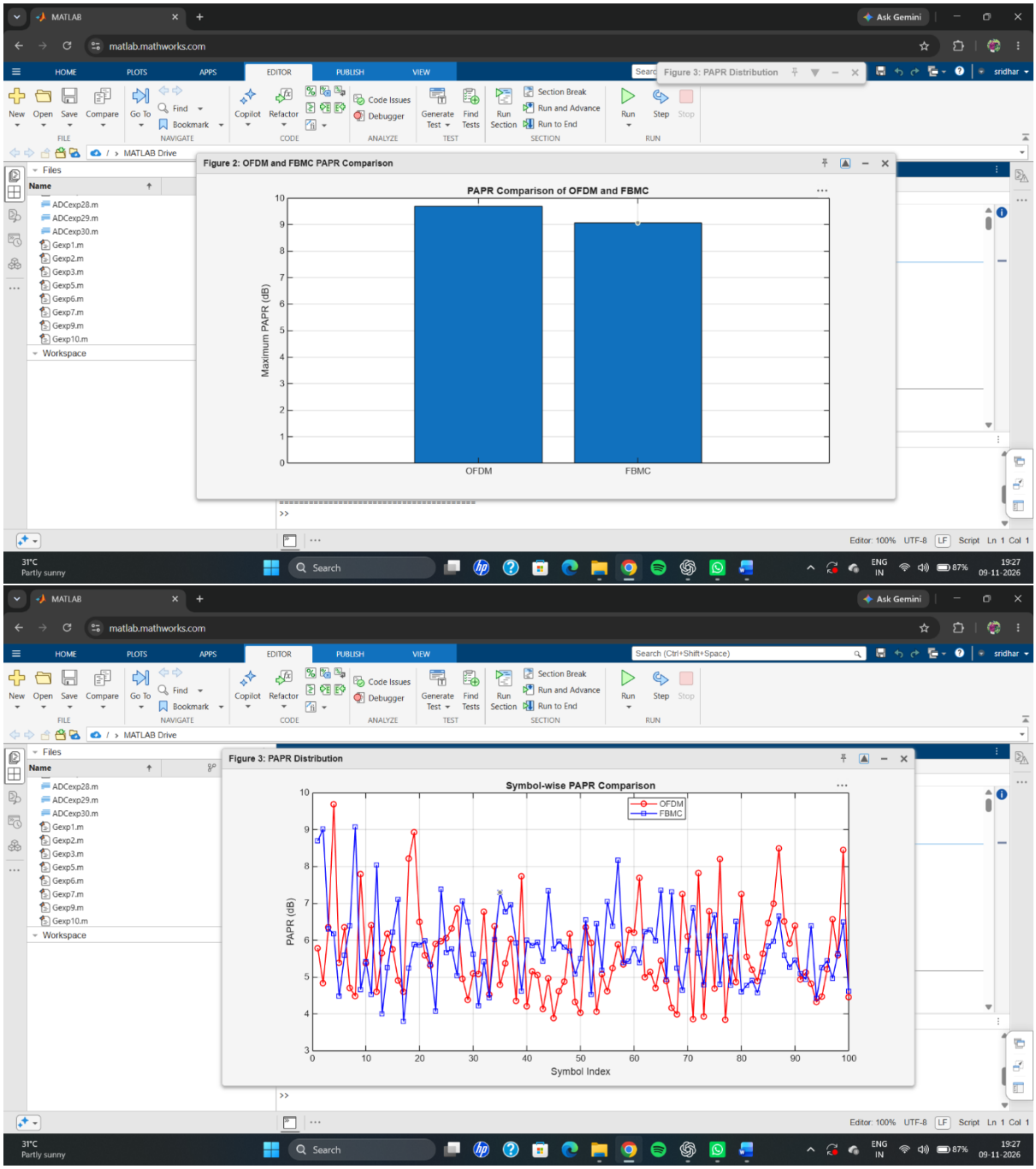
3. To examine and compare the Peak-to-Average Power Ratio (PAPR) of both waveforms under identical modulation conditions.

4. To relate the empirical results to next-generation multi-carrier waveform design constraints (GFDM, FBMC, UFMC) defined in the wireless syllabus.

PROGRAME CODE :




Figure 1: OFDM vs FBMC PSD Comparison
Power Spectral Density: OFDM vs FBMC
Normalized PSD (dB)
Normalized Frequency (cycles/sample)
OFDM
FBMC with PHYDYAS Filter



RESULT: The MATLAB simulation of OFDM and FBMC waveforms was successfully implemented using 4-QAM modulation and the PHYDYAS prototype filter. The PAPR and Power Spectral Density (PSD) of both waveforms were calculated and compared. The simulation demonstrated the spectral advantages of FBMC in reducing out-of-band emissions compared with conventional OFDM.